

EXPERIMENT – 29

Implementation of SHA-3 State Analysis for Nonzero Capacity Lanes

CODE:

```
#include <stdio.h>

int main()
{
    int blockSize = 1024;
    int stateSize = 1600;
    int capacity = stateSize - blockSize;

    int laneSize = 64;
    int rateLanes = blockSize / laneSize;
    int capacityLanes = capacity / laneSize;

    printf("SHA-3 Block Size = %d bits\n", blockSize);
    printf("State Size = %d bits\n", stateSize);
    printf("Capacity = %d bits\n", capacity);
    printf("Rate Lanes = %d\n", rateLanes);
    printf("Capacity Lanes = %d\n", capacityLanes);

    printf("\nInitially:\n");
    printf("Rate lanes contain nonzero bits = %d\n", rateLanes);
    printf("Capacity lanes contain nonzero bits = 0\n");

    /* Ignoring permutation, capacity lanes are never
       directly affected by message blocks */
}
```

```

printf("\nAfter absorbing P0, P1, P2, ... \n");

printf("Original capacity lanes remain unchanged.\n");

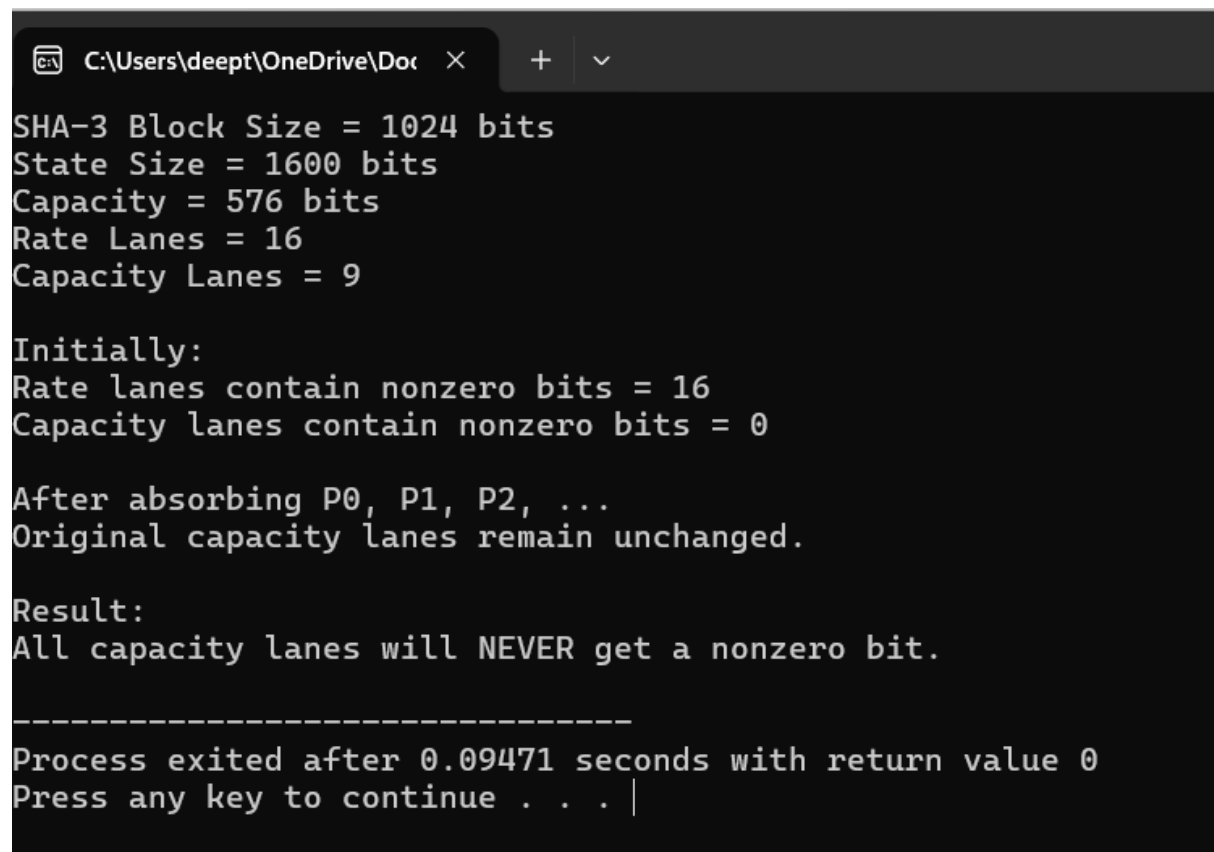

printf("\nResult:\n");

printf("All capacity lanes will NEVER get a nonzero bit.\n");


return 0;
}

```

OUTPUT:



```

C:\Users\deept\OneDrive\Doc >
SHA-3 Block Size = 1024 bits
State Size = 1600 bits
Capacity = 576 bits
Rate Lanes = 16
Capacity Lanes = 9

Initially:
Rate lanes contain nonzero bits = 16
Capacity lanes contain nonzero bits = 0

After absorbing P0, P1, P2, ...
Original capacity lanes remain unchanged.

Result:
All capacity lanes will NEVER get a nonzero bit.

-----
Process exited after 0.09471 seconds with return value 0
Press any key to continue . . . |

```